

ULANBEL, Kazakhstan

WMO: 380910

Lat: 44.800N

Lon: 71.067E

Elev: 266

StdP: 98.17

Time Zone: 6.00 (E06)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.5	-27.5	0.3	-21.8	-25.3	0.4	-20.1	10.6	-10.1	9.4	-8.1	2.8	90	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	37.7	18.4	36.0	18.0	34.5	17.5	20.7	32.7	19.5	31.6	18.6	31.1	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.2	24.9	15.0	11.0	24.3	13.6	10.0	24.3	60.8	32.4	56.6	31.8	53.6	30.8	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	7.9	DB	-24.8	41.7	4.0	1.9	-27.7	43.0	-30.0	44.2	-32.3	45.2	-35.2	46.6	(4)
(5)			WB	-25.1	24.4	4.0	2.6	-27.9	26.3	-30.2	27.8	-32.5	29.3	-35.3	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-8.0	-7.5	0.1	12.0	18.5	24.5	26.7	24.1	17.5	8.4	0.9	-5.1	(6)	
	DBStd	13.42	6.11	6.42	6.09	4.97	4.43	3.67	2.72	2.82	4.20	5.18	5.89	6.64	(7)	
	HDD10.0	2234	558	489	310	35	5	0	0	0	2	91	276	468	(8)	
	HDD18.3	3980	816	723	566	198	49	3	0	1	63	311	524	726	(9)	
	CDD10.0	2027	0	0	3	94	268	434	518	437	228	41	2	0	(10)	
	CDD18.3	729	0	0	0	7	54	188	260	179	38	3	0	0	(11)	
	CDH23.3	10020	0	0	0	87	645	2534	3691	2375	626	62	0	0	(12)	
	CDH26.7	5116	0	0	0	22	221	1329	2106	1192	232	14	0	0	(13)	
(14) Wind	WSAvg	3.1	3.0	3.7	3.7	3.7	3.3	2.9	2.9	2.9	2.9	2.7	2.7	2.9	(14)	
(15) Precipitation	PrecAvg	330	35	38	37	41	30	18	14	7	8	24	37	41	(15)	
	PrecMax	488	82	85	62	90	68	53	45	29	33	69	106	84	(16)	
	PrecMin	179	3	1	3	6	5	1	0	0	0	0	3	9	(17)	
	PrecStd	76	18	20	14	21	17	13	12	7	8	19	28	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	7.9	17.5	29.4	34.3	39.6	40.7	38.6	34.2	28.6	17.2	9.3	(19)	
		MCWB	4.5	3.6	10.7	15.7	17.3	20.0	19.5	18.4	16.8	14.6	9.8	6.0	(20)	
	2%	DB	3.9	5.0	13.5	25.6	31.2	36.8	38.3	36.2	31.4	24.7	13.8	5.7	(21)	
		MCWB	2.2	2.3	8.5	13.9	15.6	18.3	18.5	17.6	15.0	13.0	8.8	3.6	(22)	
	5%	DB	1.7	2.8	10.7	22.8	29.1	35.1	36.5	34.2	29.3	20.9	10.7	3.9	(23)	
		MCWB	0.3	0.6	6.8	12.7	15.1	17.3	18.3	17.0	14.1	10.5	6.9	2.1	(24)	
	10%	DB	0.1	0.9	8.0	20.0	27.1	33.5	34.9	32.5	27.2	17.5	8.3	2.3	(25)	
		MCWB	-0.9	-0.6	5.0	11.5	14.5	16.8	18.2	16.2	13.3	9.1	5.3	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	4.6	11.8	16.7	18.5	21.7	23.2	20.7	18.2	15.1	10.9	6.2	(27)	
		MCDB	7.0	6.8	16.2	24.1	26.9	35.7	34.9	34.2	32.4	27.4	16.5	9.2	(28)	
	2%	WB	2.2	2.6	9.3	15.2	17.2	19.7	21.1	18.9	16.1	13.2	9.2	3.8	(29)	
		MCDB	3.8	4.9	12.9	23.2	27.0	31.9	32.8	32.3	28.8	23.8	13.1	5.6	(30)	
	5%	WB	0.4	0.8	6.9	14.0	16.4	18.6	20.1	18.0	14.7	11.2	7.3	2.3	(31)	
		MCDB	1.5	2.5	10.2	21.1	26.0	30.7	31.4	31.1	27.1	19.7	10.2	3.7	(32)	
	10%	WB	-0.8	-0.6	4.8	12.3	15.5	17.9	19.1	17.1	13.5	9.5	5.3	1.2	(33)	
		MCDB	0.2	1.0	7.7	18.8	24.6	30.3	31.6	30.4	25.9	16.6	8.1	2.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	9.0	8.1	11.5	12.7	14.6	14.7	15.2	15.6	13.4	9.5	7.2	(35)	
		MCDBR	8.3	10.1	11.0	15.1	16.1	17.2	18.0	18.2	18.9	18.4	13.3	8.4	(36)	
	5% WB	MCWBR	6.7	7.7	7.0	6.5	5.9	5.6	6.0	7.1	8.1	9.4	8.8	6.4	(37)	
		MCWBR	7.9	9.4	10.8	13.0	13.3	14.8	14.9	16.0	17.1	17.4	12.6	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.349	0.383	0.432	0.467	0.466	0.451	0.442	0.407	0.379	0.392	0.383	0.349	(40)	
		taud	2.014	1.986	1.990	1.959	2.009	2.099	2.141	2.223	2.254	2.135	2.093	2.065	(41)	
	Ebn at Noon		711	764	784	793	809	823	825	843	835	752	670	667	(42)	
		Edh at Noon	108	135	154	173	170	156	148	131	117	115	100	94	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.57	2.58	3.93	5.48	6.96	7.51	7.31	6.61	5.17	3.26	1.86	1.28	(44)	
	RadStd		0.26	0.29	0.30	0.43	0.34	0.29	0.35	0.21	0.22	0.25	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page